Python Libraries for Machine Learning

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Content:

1. NumPy
2. Pandas
3. Matplotlib
4. OpenCV
5. Related Required Libraries: SciPy, Scikit-learn, XGBoost, LightGBM, Seaborn, Theano

For Deep Learning and NN: pytorch, tensorflow, keras

1. What is NumPy? NumPy is a Python library used for working with arrays.

```
Input:
import numpy
arr = numpy.array([1, 2, 3, 4, 5])
print(arr)


Output:
[1 2 3 4 5]
```

```
Input:
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print(arr)


Output:
[1 2 3 4 5]
```

```
HW: how to access array indices of 0D,
1D, 2D, 3D array?
```

```
Input:
import numpy as np

a = np.array(42)
b = np.array([1, 2, 3, 4, 5])
c = np.array([[1, 2, 3], [4, 5, 6]])
d = np.array([[[1, 2, 3], [4, 5, 6]], [[1, 2, 3], [4, 5, 6]]])

print("0D Array: ", a)
print("1D Array: ", b)
print("2D Array: ", c)
print("3D Array: ", d)

Output:
0D Array:  42
1D Array:  [1 2 3 4 5]
2D Array:  [[1 2 3]
 [4 5 6]]
3D Array:  [[[1 2 3]
  [4 5 6]]

 [[1 2 3]
  [4 5 6]]]
```

2. Pandas

Pandas is a high-level data analysis and manipulation library built on top of NumPy.

```
import pandas as pd

df = pd.read_csv("movies.csv")

print(df)
```

```
import pandas as pd

df = pd.read_excel("movies.xlsx")

print(df)
```

More functions:

```
df.duplicated()
df.head()
df.info()
```

3. Matplotlib

Matplotlib is a comprehensive data visualization library used to create static and interactive plots.

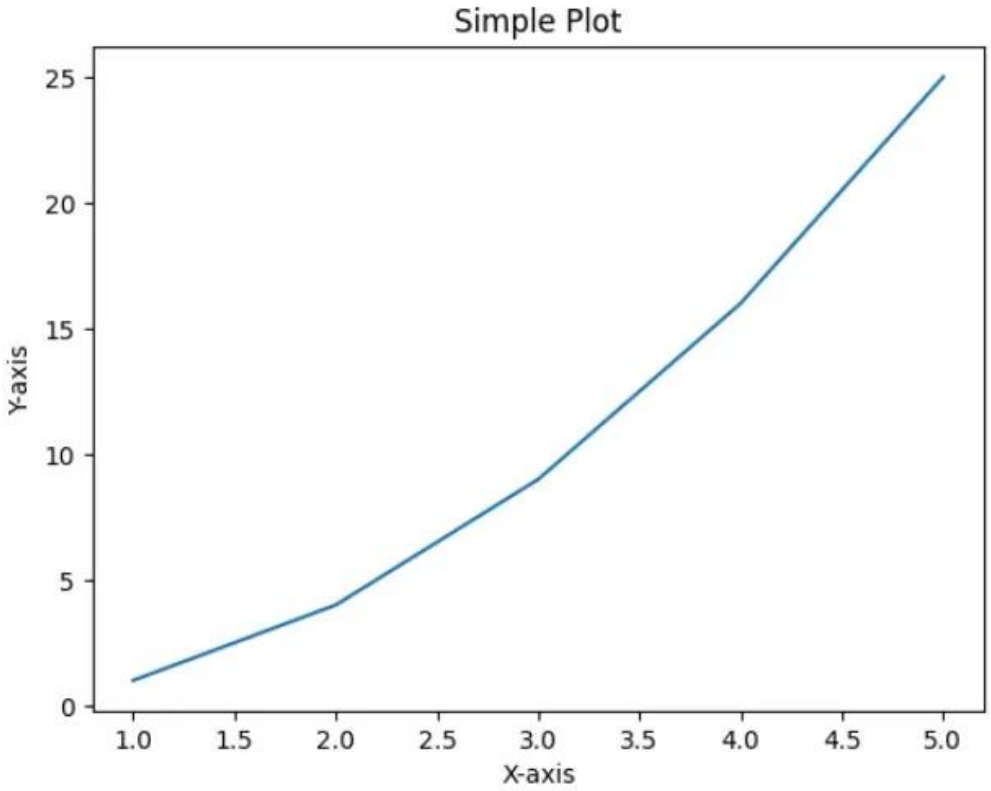
```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [1, 4, 9, 16, 25]

plt.plot(x, y)

plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Simple Plot')

plt.show()
```



H.W: Practice at home

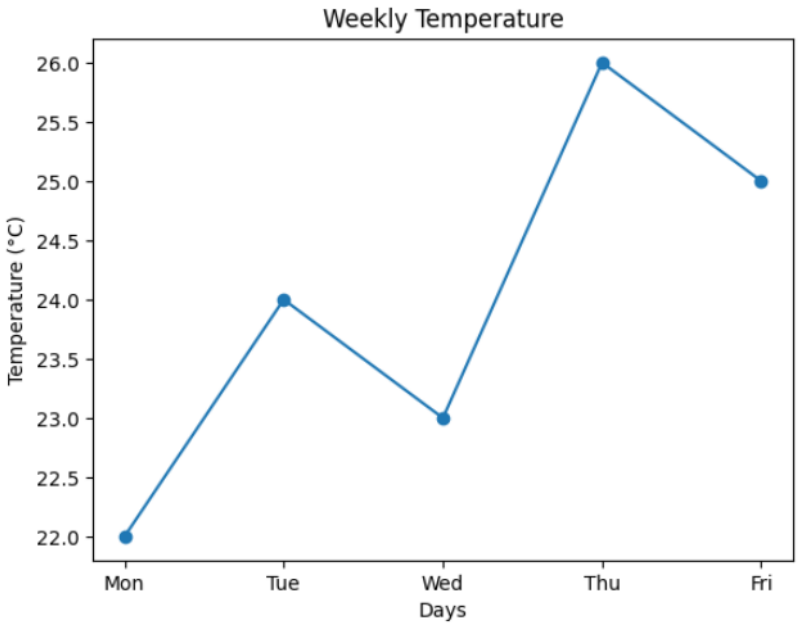
a) Line Chart
b) Bar Plot
c) Scatter Plots
d) Histograms
e) Pie Charts

a) Line chart in Matplotlib - Python

```
import matplotlib.pyplot as plt
import numpy as np

days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri']
temperature = [22, 24, 23, 26, 25]

plt.plot(days, temperature, marker='o')
plt.title('Weekly Temperature')
plt.xlabel('Days')
plt.ylabel('Temperature (°C)')
plt.show()
```

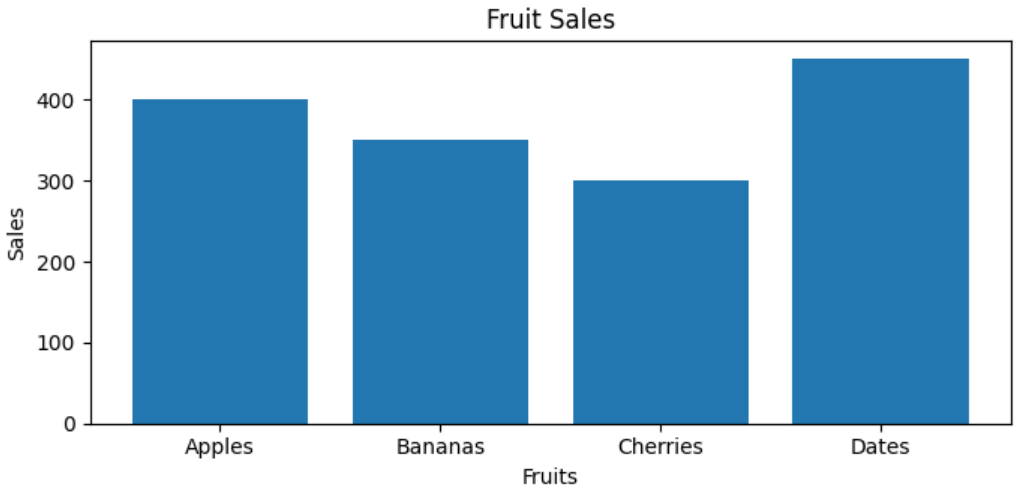


b) Bar Plot in Matplotlib

```
import matplotlib.pyplot as plt
import numpy as np

fruits = ['Apples', 'Bananas', 'Cherries', 'Dates']
sales = [400, 350, 300, 450]

plt.bar(fruits, sales)
plt.title('Fruit Sales')
plt.xlabel('Fruits')
plt.ylabel('Sales')
plt.show()
```

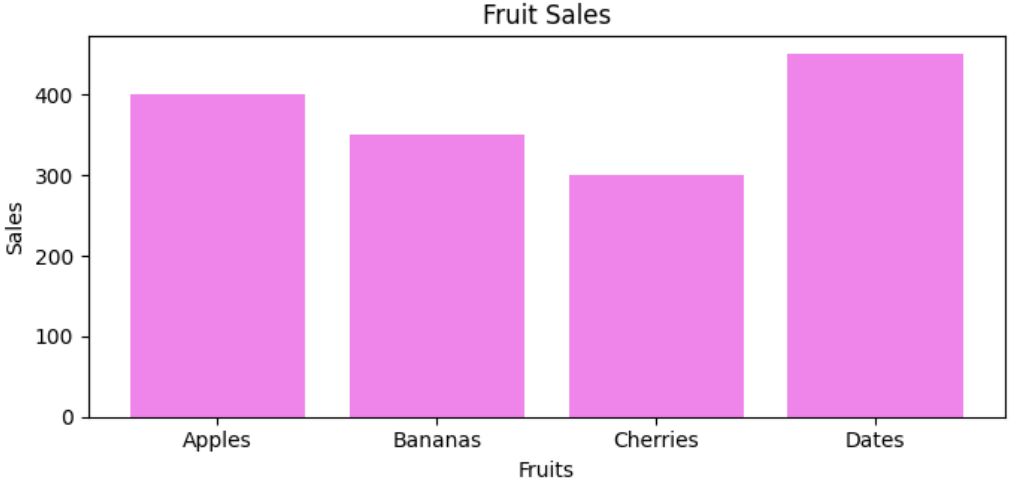


Custom color

```
import matplotlib.pyplot as plt
import numpy as np

fruits = ['Apples', 'Bananas', 'Cherries', 'Dates']
sales = [400, 350, 300, 450]

plt.bar(fruits, sales, color='violet')
plt.title('Fruit Sales')
plt.xlabel('Fruits')
plt.ylabel('Sales')
plt.show()
```



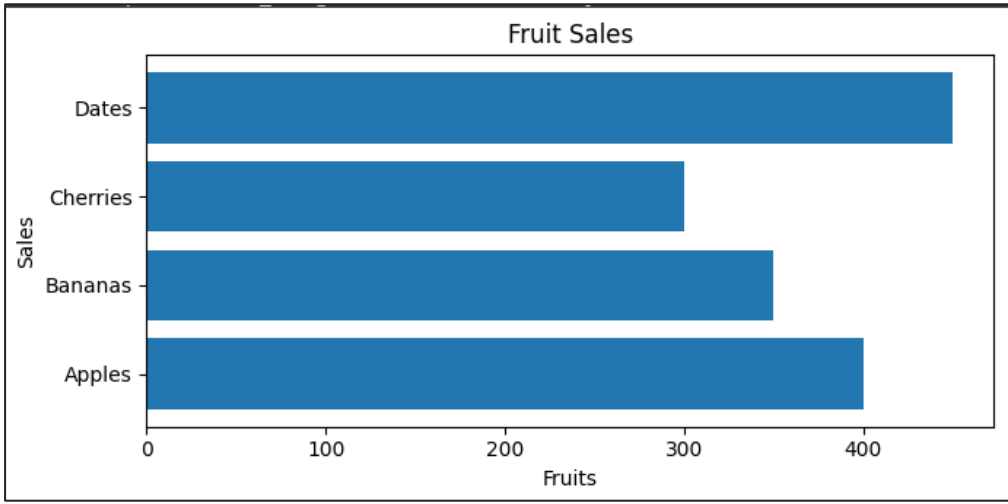
Creating Horizontal Bar Plots

```
import matplotlib.pyplot as plt
import numpy as np

fruits = ['Apples', 'Bananas', 'Cherries', 'Dates']
sales = [400, 350, 300, 450]

plt.barh(fruits, sales)

plt.title('Fruit Sales')
plt.xlabel('Fruits')
plt.ylabel('Sales')
plt.show()
```

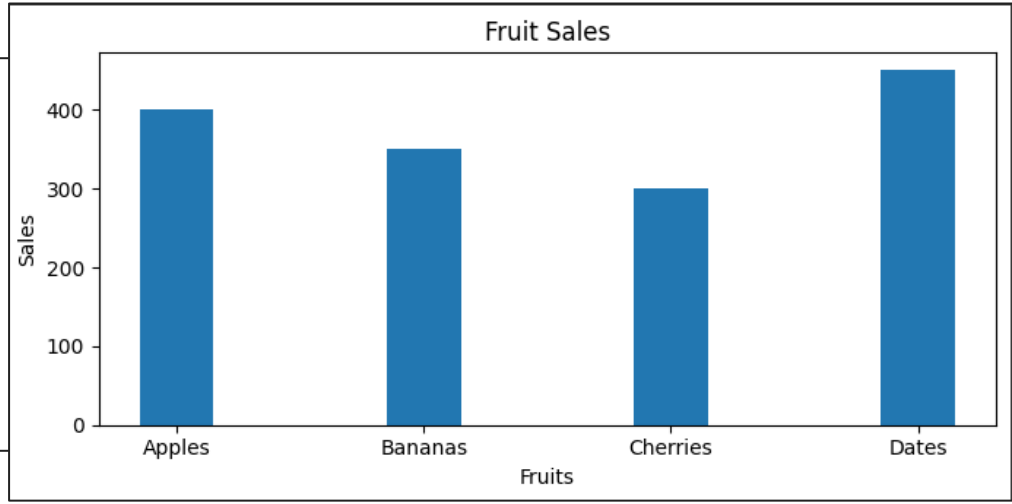


Adjusting Bar Width

```
import matplotlib.pyplot as plt
import numpy as np

fruits = ['Apples', 'Bananas', 'Cherries', 'Dates']
sales = [400, 350, 300, 450]

plt.bar(fruits, sales, width=0.3)
plt.title('Fruit Sales')
plt.xlabel('Fruits')
plt.ylabel('Sales')
plt.show()
```

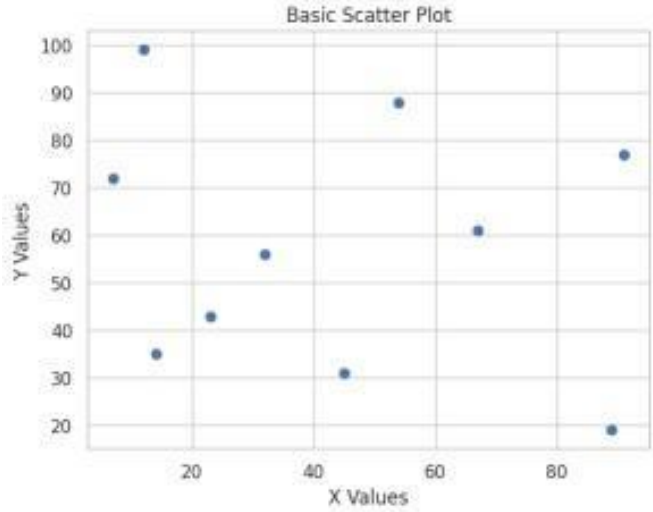


c) Matplotlib Scatter

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([12, 45, 7, 32, 89, 54, 23, 67, 14, 91])
y = np.array([99, 31, 72, 56, 19, 88, 43, 61, 35, 77])

plt.scatter(x, y)
plt.title("Basic Scatter Plot")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.show()
```



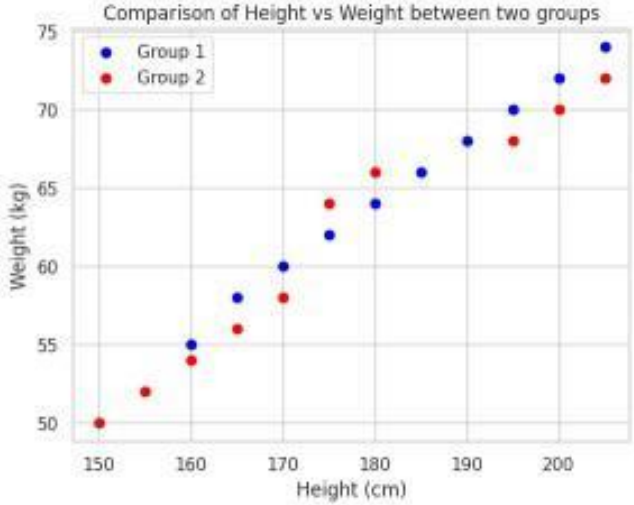
Two group scatter

```
x1 = np.array([160, 165, 170, 175, 180, 185, 190, 195, 200, 205])
y1 = np.array([55, 58, 60, 62, 64, 66, 68, 70, 72, 74])

x2 = np.array([150, 155, 160, 165, 170, 175, 180, 195, 200, 205])
y2 = np.array([50, 52, 54, 56, 58, 64, 66, 68, 70, 72])

plt.scatter(x1, y1, color='blue', label='Group 1')
plt.scatter(x2, y2, color='red', label='Group 2')

plt.xlabel('Height (cm)')
plt.ylabel('Weight (kg)')
plt.title('Comparison of Height vs Weight between two groups')
plt.legend()
plt.show()
```



d) Plotting Histogram in Python using Matplotlib

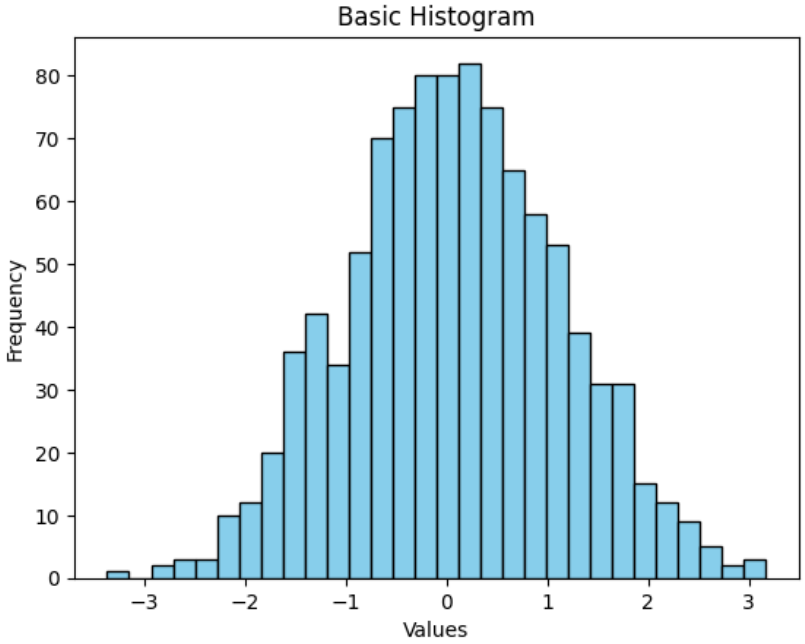
```
import matplotlib.pyplot as plt
import numpy as np

data = np.random.randn(1000)

plt.hist(data, bins=30, color='skyblue', edgecolor='black')

plt.xlabel('Values')
plt.ylabel('Frequency')
plt.title('Basic Histogram')

plt.show()
```



e) Plot a Pie Chart in Python using Matplotlib

Creating Pie Charts

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])

plt.pie(y)
plt.show()
```

Labels

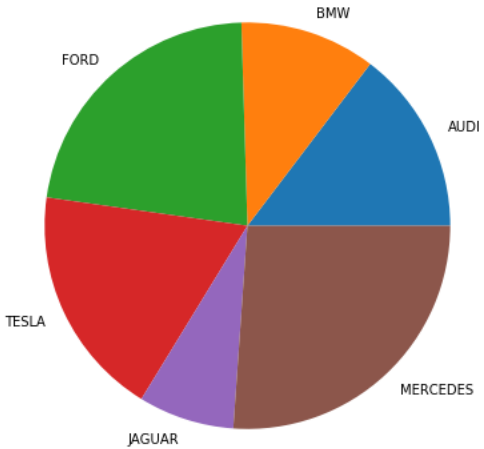
```
from matplotlib import pyplot as plt
import numpy as np


cars = ['AUDI', 'BMW', 'FORD', 'TESLA', 'JAGUAR', 'MERCEDES']

data = [23, 17, 35, 29, 12, 41]

fig = plt.figure(figsize=(10, 7))
plt.pie(data, labels=cars)

plt.show()
```

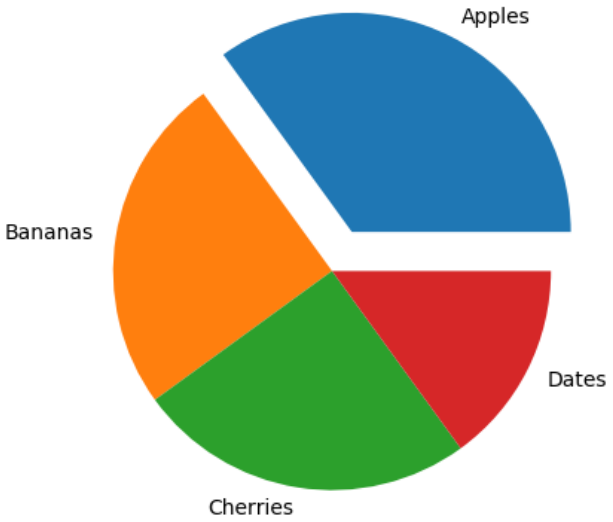


Shadow

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels = ["Apples", "Bananas", "Cherries", "Dates"]
myexplode = [0.2, 0, 0, 0]

plt.pie(y, labels = mylabels, explode = myexplode)
plt.show()
```

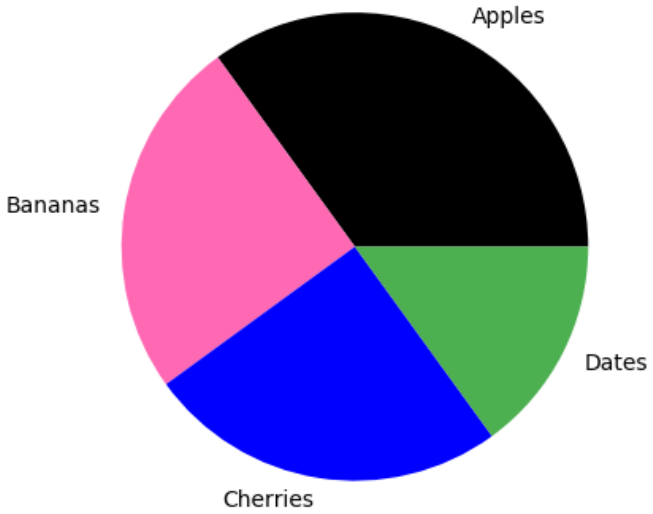


Colors

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels = ["Apples", "Bananas", "Cherries", "Dates"]
mycolors = ["black", "hotpink", "b", "#4CAF50"]

plt.pie(y, labels = mylabels, colors = mycolors)
plt.show()
```

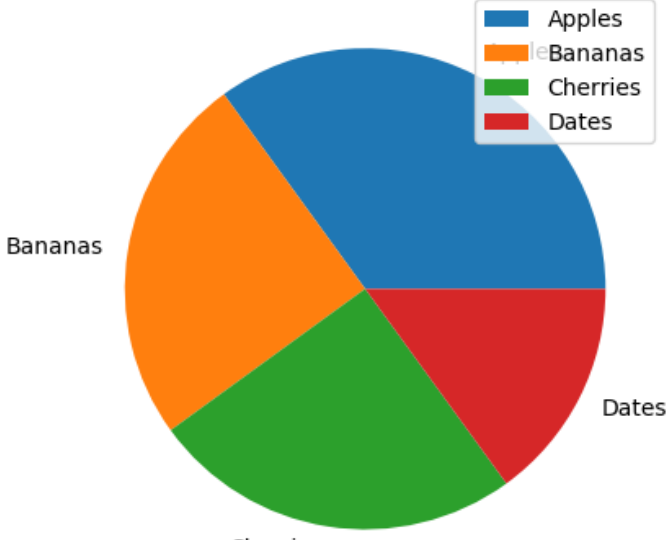


Use of Legend

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels = ["Apples", "Bananas", "Cherries", "Dates"]

plt.pie(y, labels = mylabels)
plt.legend()
plt.show()
```

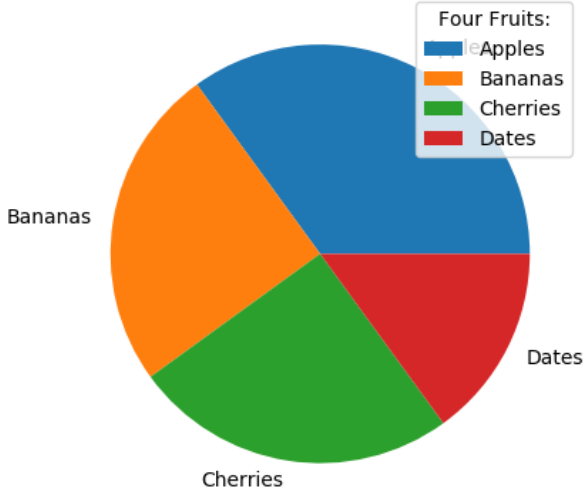


Legend With Header

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels = ["Apples", "Bananas", "Cherries", "Dates"]

plt.pie(y, labels = mylabels)
plt.legend(title = "Four Fruits:")
plt.show()
```



❑Introduction to OpenCV

Display image:

```
import cv2
img = cv2.imread('your_image.jpg')
cv2.imshow('My Image Window', img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Display webcam:

```
import cv2

cap = cv2.VideoCapture(0)
if not cap.isOpened():
    print("Error: Could not open webcam.")
    exit()

print("Camera opened successfully. Press 'q' to quit.")
while True:
    ret, frame = cap.read()
    if not ret:
        print("Error: Failed to grab frame.")
        break

    cv2.imshow('My Webcam Feed', frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

H.W: Basic Computer Vision projects using opencv and mediapipe

See You in the Next Lecture...